



Seeing only what we can see: why cyber strategy may be built on illusion

Francesco Ferazza^{1,2}

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Abstract

Debates about cyber strategy lean on a handful of high-profile operations—Stuxnet, NotPetya, SolarWinds—as if they were clean tests of competing theories. But the evidence we get to see is filtered by actors with their own reasons for revealing or concealing, and that filtering is not neutral. We call the result a “visibility trap”: paradigms whose claims happen to predict the kind of evidence the public record produces look well supported, while paradigms whose claims turn on what stays hidden—coercion above all—struggle to find a clean test even where they may be right. We develop two concepts, the “visibility pipeline” and “epistemic congruence,” to trace how clandestine operations become public knowledge and to show why some theories are advantaged at the point of evidence rather than at the point of truth. The argument is epistemic, not empirical: we do not claim the dominant paradigms are wrong, only that we cannot tell from public evidence whether they dominate because they are correct or merely because they are visible. Recognising the trap is a first step toward a more honest, and more useful, account of cyber strategy.

Keywords Cybersecurity · Offensive cyber operations · Epistemology · Strategy · Security · Methodology

1 Introduction

Strategic knowledge about cyberspace is built under unusual constraints. Most offensive cyber operations are clandestine: only a fraction are ever detected, fewer are publicly attributed, and almost none are documented in a way that reveals what the operators were actually trying to achieve. And yet our debates lean on a small set of well-publicised operations—Stuxnet, NotPetya, SolarWinds—as though they were transparent windows onto strategy.

Take NotPetya. On 27 June 2017 it tore far beyond its apparent Ukrainian targets and did billions of dollars in damage worldwide, and it quickly became a fixture in arguments about what cyber operations can do and whether anyone can keep them under control. Stuxnet (2010) gets cited as proof that code can stand in for kinetic force; SolarWinds (2020) as the emblem of permanent, low-level competition. Each case carries a lot of theoretical weight.

The argument of this paper is that these operations became paradigm-defining less because they *settled* anything than because they passed through a *visibility pipeline* that makes some kinds of evidence abundant and others almost impossible to find. The effect is a quiet bias toward whichever paradigm happens to ask for the evidence that vendors, governments, and victims are willing and able to produce.

The core claim is that cyber strategic paradigms differ in their **epistemic congruence**—how well the evidence a theory needs matches the evidence the pipeline actually yields. High-congruence paradigms (institutional critique, competition) pile up confirming cases; low-congruence ones (coercion) can describe real behaviour and still find no clean test. The intuition is an old one in strategic studies: successful deterrence leaves few footprints, so the very operations that would vindicate a coercion paradigm tend to be the ones that leave nothing for the analyst to see. This is the *visibility trap*.

Contribution. This is a theory-building paper, not an empirical verdict. We develop two concepts—the visibility pipeline and epistemic congruence—that generate testable hypotheses, trace the mechanisms linking congruence to paradigm dominance, and set out scope conditions. The cases that

✉ Francesco Ferazza
Francesco.Ferazza.2021@live.rhul.ac.uk

¹ Royal Holloway, University of London, London, UK

² Marina Militare, Rome, Italy

follow are there to illustrate the framework and show it doing work; they are not meant to settle the contest between paradigms.

What this adds to existing debates is worth stating plainly, because the difficulty of cyber coercion is not itself a new observation. Borghard and Loneragan show that cyber operations rarely meet the classical requisites of coercion—a clearly communicated threat, credibility, reassurance—and conclude that cyber power alone makes a weak coercive instrument [2]. Nye argues that the nuclear analogy misleads, and that prevention in cyberspace runs through punishment, denial, entanglement, and norms rather than retaliatory threats [34]. These are *substantive* claims about how cyberspace works. Ours is an *epistemic* one: from public evidence we cannot tell whether coercion performs badly because it is genuinely weak or because its successes are invisible by construction. The familiar conclusion that cyber coercion “does not work” may be partly an artifact of the same pipeline we describe—one that manufactures the visible failures the conclusion rests on while quietly absorbing the compliance that would complicate it.

This also marks the paper off from its nearest relative. Lupovici diagnoses that secrecy leaves scholars unable to identify cyber-deterrence success *or* failure, and proposes studying the *adoption* of a deterrence strategy rather than its outcomes [26]. We push the point in two directions. His problem sits *within* one paradigm—how to study deterrence when it is invisible; ours is a bias *between* paradigms, where the pipeline does not just hide one paradigm’s outcomes but tilts the whole field, lending the visible frameworks the look of having won the argument. And his remedy does not get clear of the trap, since “adoption” is read off doctrine and attribution statements that are themselves pipeline products. The visibility trap, on my account, is not a problem you solve by moving the dependent variable; it is built into how cyber strategic knowledge gets made.

Theoretical positioning. The visibility trap framework draws on scholarship largely absent from cyber strategy debates. First, it extends *agnotology*—the study of culturally produced ignorance—to security studies [35]. Agnotology asks not only how we know but why we *do not* know: how institutional processes generate structured blind spots rather than merely incomplete information. The visibility pipeline is an agnotological mechanism, systematically producing ignorance about operations that do not cross detection thresholds or serve attribution interests.

Second, the framework engages with the political economy of expertise. Haas’s work on *epistemic communities* shows how networks of knowledge-based experts control the framing of policy problems and acceptable solutions [14]. Jasanoff’s co-productionist framework demonstrates that scientific knowledge and social order are mutually constitutive

[15]. In cyber strategy, vendors, governments, and academic institutions collectively constitute an epistemic regime whose outputs we treat as neutral evidence rather than artifacts of particular interests and capabilities.

Third, the framework builds on methodological debates in intelligence studies about the epistemology of secrecy. Scholars of covert action have long recognized that studying secret state behavior through public sources produces systematically distorted knowledge. Trevorton observed that covert action “failures, such as the Bay of Pigs, are apparent, but successes are harder to judge” [43]—a selection bias that shapes what theories can be tested. Carson’s distinction between what states *know* and what they *acknowledge* reveals that covert operations often exist as “open secrets”—known to adversary intelligence services but invisible to publics and scholars [4]. Cormac and Aldrich’s concept of “implausible deniability” highlights how visibility operates on a spectrum across different audiences [6]. The visibility trap extends these insights to the cyber domain, where vendor-mediated technical forensics creates an additional layer of epistemic infrastructure absent in traditional covert action.

Paper structure. Section 2 develops the theoretical framework. Section 3 maps three paradigms onto this framework. Section 4 outlines methodology. Sections 5, 6 and 7 analyze Stuxnet, NotPetya, and SolarWinds. Section 8 examines negative cases. Section 9 synthesizes findings. Section 10 concludes.

2 Theory: the visibility pipeline as epistemic infrastructure

2.1 The problem: strategic knowledge under secrecy

Cyber operations are hard to study because the analytically relevant variables—intent, decision-making, alternative options, cost-risk assessments—are hidden. What we observe instead is trace evidence: malware samples, forensics, victim reports, patch cycles, attribution statements. These artifacts are downstream products of operational choices, detection thresholds, and institutional incentives—not transparent windows into strategy.

Research on covert action has long grappled with studying secret behavior through filtered public traces [4, 6]. The cyber domain exacerbates this challenge: the technical layer is unusually salient to specialized vendors, while strategic context remains classified. Most of what outsiders can confidently say about major operations concerns tactics, tools, and victimology—not the strategic calculus behind them.

This creates a **structural asymmetry**: scholars and doctrine writers have abundant access to certain evidence types and almost none to others. Paradigms that build on observ-

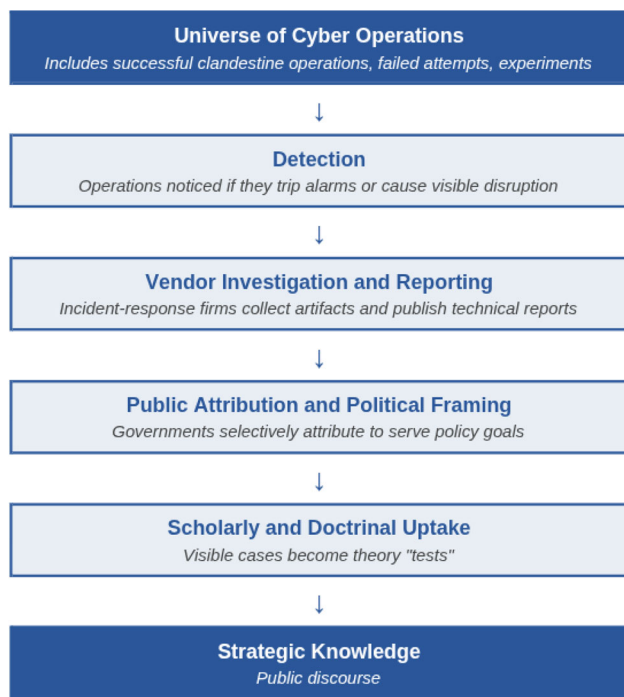


Fig. 1 The visibility pipeline: from operations to public knowledge

able technical patterns enjoy structural advantages over those requiring insight into classified decision-making.

2.2 Mechanism: the visibility pipeline

The *visibility pipeline* is the process through which clandestine operations become public strategic knowledge:

At each stage, actors apply selection criteria orthogonal to strategic inference (Fig. 1). Detection favors noisy or technically unusual operations. Vendors prioritize cases demonstrating their expertise. Governments attribute when it serves policy. Scholars gravitate toward well-documented cases fitting recognized debates.

2.3 Scope conditions

The visibility trap should be strongest when: (1) intent and strategic context are highly classified, (2) technical artifacts are rich and widely circulated, and (3) significant contestation exists over the meaning of operations. It is weaker where declassification, public hearings, or leaks make intent easier to assess.

Although this paper focuses on states, the underlying concealment dynamic is not state-specific, and recognizing this strengthens rather than narrows the argument. Private firms exhibit the same incentive structure: banks and other targets are routinely reluctant to disclose that they have suffered a cyber-attack, fearing reputational and market consequences, so the public record of incidents is filtered by commer-

cial as well as governmental interests. This matters for our framework in two ways. First, it confirms that the pipeline's distortions arise from generic incentives to manage disclosure rather than from anything peculiar to national security. Second, commercial actors are not merely additional concealers outside the pipeline; the vendors who produce the authoritative technical record are themselves private firms with disclosure incentives of their own. We retain a state focus because the strategic paradigms under examination are theories of state behavior, and because the stakes of doctrinal misperception are highest there; but the mechanism generalizes, and the financial sector offers a natural site for testing it.

2.4 Structured ignorance, not mere absence

The visibility pipeline does not simply filter information; it actively produces structured ignorance. Ignorance as absence suggests a gap that more research might fill. Structured ignorance, by contrast, has institutional foundations: it is maintained by the same processes that generate what we do know [35].

This parallels Carson's finding that covert military interventions are "almost always detected by other major powers," yet both intervener and target often maintain secrecy through "tacit collusion" [4]. Carson substantiates this claim through cases of limited war and covert intervention—among them the Spanish Civil War, the Korean War, and Vietnam—in which great powers possessed the intelligence apparatus to detect one another's covert participation but shared an interest in keeping that participation a backstage "open secret" in order to limit escalation. Detection, on this account, is not the same as acknowledgement: rival states often know, while publics and outside observers do not. The concealment operates primarily between governments and domestic publics—and, crucially, between governments and researchers. Adversary intelligence services likely know far more about cyber operations than the public record reflects; the visibility pipeline produces ignorance specifically for scholars and publics, not for the states actually engaged in competition.

Two features of the cyber domain qualify the analogy, and both sharpen rather than weaken the argument. First, Carson's detecting actor is a rival *state* intelligence service; in cyberspace the actor that first detects and characterizes an operation is frequently a commercial *vendor*, whose incentives to publish (or not) differ from those of a government. Second, attribution in cyberspace is genuinely more uncertain than the detection of troops or aircraft, so the gap between "known" and "acknowledged" is mediated by a private technical-forensics industry that has no analogue in Carson's cases. The cyber "open secret" therefore has a three-way structure—state, vendor, public—rather than Carson's

two-way structure of intervener and rival. This intervening commercial layer is precisely the epistemic infrastructure the visibility trap is meant to capture. Crucially, the vendor is not merely a passive conduit for a dynamic the states control. Where Carson's intervener and rival share an incentive to keep an operation backstage to manage escalation, the commercial vendor has a competing incentive to pull artifacts *onstage*—publication demonstrates technical authority and market position—and that incentive is largely indifferent to the principals' escalation concerns. The vendor layer can therefore *disrupt* state collusion, forcing public exposure before either government is ready to acknowledge an incident. Visibility in cyberspace is thus not only mediated but sometimes actively driven by an actor whose interests are orthogonal to the strategic stakes, which is itself a core reason the public record is a poor sample of what states are doing.

In cyber strategy, structured ignorance takes specific forms. We lack systematic knowledge of successful clandestine operations—not because no one has looked, but because the institutional apparatus for producing public knowledge is designed to surface failures and anomalies. We lack insight into coercive signaling dynamics—not because they are inherently unobservable, but because the actors who control evidence production have no incentive to document them.

A disclosure from 2026 illustrates how long this ignorance can persist—and how the pipeline shapes not only *whether* an operation becomes known but *when*. In April 2026, researchers at SentinelOne reported on “fast16,” a modular sabotage framework whose components carry 2005 compilation timestamps and which they assess was designed to corrupt the outputs of high-precision engineering and physics simulations rather than to damage equipment directly [18]. Symantec's threat-hunting team subsequently corroborated the analysis, identifying the targeted software and a logic tied to the simulation of nuclear-implosion physics [41].

If these assessments hold, the operation would predate the publicly known Stuxnet timeline, and would have pursued broadly similar objectives by different means. What makes the case relevant here is less the artifact than its chronology of *discovery*: the principal sample was uploaded to a public malware repository roughly a decade ago—scanned by dozens of commercial engines yet flagged by virtually none—and was only analysed and published in April 2026, some twenty years after the 2005 build. This dwell time is worth dwelling on, because it is the part of the story that requires no trust in any vendor's analytical judgement: independent of what fast16 was or whom it targeted, the bare fact that a sophisticated sample could remain openly available and effectively invisible for ten years is an observable property of the public record itself.

The case should be handled with care, and that caution is itself instructive. The reporting is recent and rests largely on

two vendor analyses; attribution and the identity of the target remain matters of assessment rather than confirmed fact; and the operation's relationship to Stuxnet is openly contested, with informed observers placing it anywhere from five years before Stuxnet to roughly contemporaneous with its development [48]. We do not rely on fast16 to establish any empirical claim about the history of cyber operations. Its value for our argument is the opposite: that even with the sample in hand, the community cannot yet agree on when the operation occurred or whom it targeted is a vivid demonstration that the public chronology of cyber conflict is constructed and provisional, lagging deployment by years and remaining contested long after disclosure. The visibility pipeline does not merely hide operations; it releases them late, partially, and into dispute. The irony is not lost on us, and it is worth stating plainly: fast16 reaches us only through the very vendor-mediated channel this paper treats as an interested producer of knowledge. We read it accordingly—not as settled fact about 2005, but as a specimen of how such facts are manufactured, delayed, and contested. That a claim of this magnitude enters the record through two competing security firms, is recruited by at least one commentator as evidence of national cyber prowess [1], and is disputed almost immediately on its central chronological claim is not a complication for our argument but a live enactment of it.

fast16 is the most striking instance of a recurring pattern rather than an isolated one. The lag between deployment and public understanding is visible across the canonical cases: an early Stuxnet variant active around 2007 was only disclosed in 2013 [30], and the “Equation Group” tooling whose activity reaches back to roughly 2001 was not characterized publicly until 2015 [19]. Each interval marks a span during which scholarship and doctrine necessarily reasoned from an incomplete record—and could not know how incomplete it was.

2.5 Three sources of epistemic distortion

1. Selection effects. The public record overrepresents operations that fail (discovered because they malfunctioned or caused spillover), operations against high-value monitored targets, and operations whose discovery served political purposes. It underrepresents successful clandestine operations and those quietly remediated.

2. Metric bias. Success tends to be defined by what is measurable: technical persistence can be quantified; strategic leverage cannot. Spillover is apparent; quietly deterred behavior is unseen.

3. Authority bias. Technical forensics substitutes for strategic analysis. Vendors hold epistemic authority over what transpired technically, and their reports become the de facto record—even when saying nothing about intent.

3 Three paradigms and their epistemic congruence

Having established the visibility pipeline as epistemic infrastructure, we now examine how it differentially advantages competing strategic paradigms. Three dominant paradigms structure cyber strategic thought: *coercion*, *institutional critique*, and *competition*. Each implies different evidentiary requirements. The visibility pipeline systematically favors paradigms whose requirements align with what vendors and governments can observe and publish.

3.1 The coercion paradigm (c. 2005–2012)

Core claims: Cyber operations can function as coercive instruments: by imposing or threatening costly disruption, states can compel or deter adversaries [23, 33].

Epistemic requirements: Explicit or implicit threats/demands; calibrated effects interpretable as punishment; clear attribution; observable target behavior change.

Epistemic congruence: LOW. Effective signaling requires remaining both operationally effective and publicly legible, yet practices maximizing effectiveness (stealth, deniability) make signals harder to interpret [11]. From public evidence, observers cannot distinguish operations meant to coerce from those pursuing other objectives. Coercion may perform poorly empirically *and* suffer low congruence; the visibility pipeline collapses these distinct problems. The coercion paradigm's low congruence is not an incidental feature of cyberspace but a direct consequence of how coercion works, and the classical strategic-studies literature supplies the mechanism. Schelling's distinction between *deterrence* and *compellence* is decisive here [38]. Deterrence—dissuading an adversary from acting—succeeds as a *non-event*: when it works, nothing happens, the demand can remain implicit, and there is no episode for the analyst to observe. Compellence—making an adversary actively change course—by contrast requires a communicated demand, typically a visible deadline, and an observable act of compliance. The two halves of coercion therefore have sharply different observability, and the deterrent half, on which much cyber coercion would rest, is close to invisible by construction.

This is the methodological problem that Lebow and Stein identified for deterrence theory more broadly: deterrence success is an “elusive dependent variable” because the absence of an attack is consistent both with successful deterrence and with the adversary never having intended to act [22]. Successful deterrence, in the familiar formulation, leaves few footprints. The visibility pipeline does not create this problem; it inherits it from the structure of coercion and then compounds it, because the actors who control cyber evidence

production have no incentive to document the quiet compliance that would constitute a deterrent success.

The observability of compellence should not be overstated, and this is where the cyber case adds a twist to the classical picture. In principle compellence is the legible half of coercion: an explicit demand, a deadline, an observable act of compliance. But legibility here is a matter of degree, not a guarantee. Even a genuinely compelling cyber operation can be stripped of its political signal by the pipeline, because the *technical* layer of a coercive intrusion is often indistinguishable from routine intelligence collection or from a sabotage attempt that simply failed. Absent an unusually explicit and publicly broadcast demand, the artifact that reaches analysts—tooling, access, payload—underdetermines whether anyone was being coerced at all. This refines rather than contradicts the deterrence–compellence asymmetry: compellence remains *relatively* more visible than deterrence, which is why the rare cases that surface (Sony, discussed below) are compelling ones, but the pipeline degrades even that relative advantage, so the absence of a visible cyber-coercion record cannot be read as evidence that coercion is not occurring.

The signaling requirement deepens the difficulty. For a cyber operation to coerce, its target must read it *as* a signal and correctly infer the sender and the demand. Jervis's distinction between *signals*—moves whose meaning is conventional and therefore manipulable—and *indices*—features taken to be inherently linked to capability or intent—clarifies why this is so hard in cyberspace [16]. The properties that make a cyber operation strategically useful (deniability, stealth, ambiguity of effect) are precisely the properties that destroy its legibility as a signal, leaving the target unable to distinguish a deliberate “calling card” from careless tradecraft. And because perception is theory-driven rather than neutral [17], observers tend to assimilate ambiguous operations to whichever paradigm already dominates—reinforcing, at the level of cognition, the selection bias the pipeline imposes at the level of evidence.

3.2 The institutional critique (c. 2012–2018)

Core claims: Cyber operations are constrained by organizational frictions, legal processes, and political risk-aversion. Offense is harder than it appears; operational control is fragile [36, 39].

PETIO framework [40]: High-impact operations require People, Exploits, Toolset, Infrastructure, and Organizational structure—making large-scale, repeatable operations rare and fragile.

Trilemma: Speed, intensity and control are negatively correlated in subversive operations; a gain in one tends to produce losses in the other two [28]. A related dynamic is capability

burn: discovery degrades the tools and access an operation relied on.

Epistemic congruence: HIGH. Vendor reports are rich in precisely the artifacts (tooling, timelines, spillover) that PETIO and trilemma arguments highlight.

3.3 The competition paradigm (c. 2016–present)

Core claims: Cyber operations constitute continuous campaigns to shape the information environment, gain intelligence, and secure positional advantage below the threshold of war [5, 9, 29, 32].

Epistemic congruence: MEDIUM–HIGH. Vendor reports increasingly emphasize long-running campaigns and persistent presence. But how positional advantages translate into leverage remains classified or speculative.

A prominent variant of this paradigm reframes cyber competition as an *intelligence contest* rather than warfare or coercion: an effort to steal and operationalize secrets for relative advantage, exhibiting the characteristics of traditional espionage rather than of compellent force [37]. This framing is analytically powerful, but for our purposes it is also instructive as an object lesson in congruence: espionage-style competition predicts exactly the observables the pipeline produces best—dwell time, stealth, exfiltration, persistent access—which is part of why an ambiguous operation such as SolarWinds was so readily assimilated to it. The intelligence-contest reading may well be correct; our point is that its evidentiary requirements are unusually well matched to what vendors can document, so its apparent confirmation cannot by itself settle whether it is correct or merely congruent.

3.4 Observational equivalence

The visibility pipeline creates an observational equivalence problem. When a paradigm has high epistemic congruence, the public record looks similar whether that paradigm is substantively correct or merely pipeline-congruent. Table 1 summarizes the relationship between paradigms and their epistemic requirements.

From public evidence alone, observers cannot tell whether a paradigm's congruence and validity have diverged. This is the core of the visibility trap.

4 Methodology: cases as epistemic events

4.1 Research design and objectives

This paper pursues theory-building rather than theory-testing. The goal is not to adjudicate definitively among competing cyber strategy paradigms, but to develop conceptual

tools—the visibility pipeline and epistemic congruence—that illuminate how the field's evidentiary base shapes theoretical development. Theory-building research of this kind aims to generate new concepts, identify causal mechanisms, and produce testable hypotheses for subsequent empirical work [12].

The analysis proceeds through structured comparison of canonical cases, treating each not as a transparent test of strategic theory but as an *epistemic event*: an incident whose passage through the visibility pipeline reshaped what counts as credible strategic knowledge. This approach draws on constructivist methodology in security studies, which examines how shared understandings emerge and stabilize through discourse and institutional practice [46]. It also builds on recent work addressing the epistemic limits of studying covert action through public traces [24].

4.2 Case selection logic

Case selection follows the logic of theory-building from paradigmatic cases rather than representative sampling [10]. Stuxnet (2010), NotPetya (2017), and SolarWinds (2020) were chosen on three criteria:

First, **paradigm-defining status**: each case is widely cited as empirical support for particular theoretical positions. Stuxnet anchors institutional critique arguments about PETIO burdens and capability burn; NotPetya is invoked as evidence for trilemma dynamics; SolarWinds exemplifies competition-paradigm claims about persistent engagement. Selecting cases that *already function as paradigm anchors* allows us to examine how they achieved that status—precisely the question the visibility trap framework addresses.

Second, **documentation richness**: each case has unusually extensive public documentation, including vendor technical reports, government attribution statements, journalistic investigations, and academic analysis. This richness is itself a product of the visibility pipeline, making these cases particularly suitable for tracing how evidence production shapes interpretation.

Third, **variation across operation types**: together, the cases span sabotage (Stuxnet), destructive attack (NotPetya), and espionage (SolarWinds), allowing examination of whether the visibility trap operates consistently across different operational categories.

This is not a most-similar or most-different design aimed at isolating causal variables. Rather, it is an analytic strategy for theory development: using well-documented cases to elaborate the visibility trap mechanism and identify its scope conditions.

Table 1 Epistemic congruence across paradigms

Paradigm	Key claims	What must be observable	Congruence
Coercion	Operations coerce through signaling and punishment	Communicated intent / explicit demands, target updating	LOW
Institutional critique	Operations face PETIO burdens and trilemma trade-offs	Sophistication, spillover, capability burn	HIGH
Competition	Operations yield positional advantages in ongoing campaigns	Stealth, persistence, dwell time	MED–HIGH

4.3 Negative cases

Section 8 examines WannaCry (2017) and Sony Pictures (2014) as negative cases—incidents with high visibility that nonetheless failed to achieve paradigm-defining status. Negative case analysis is essential to theory-building because it identifies boundary conditions and prevents over-generalization [27]. If the visibility trap framework claimed that visibility alone determines paradigm influence, these cases would constitute disconfirming evidence. Their inclusion allows refinement of the framework by specifying mediating factors (attribution frames, target characteristics, domain boundaries) that condition the trap's operation.

4.4 Analytic strategy

For each case, the analysis proceeds through four steps:

1. **Visibility story:** How was the operation detected, investigated, attributed, and framed? Which actors produced the public record, and what were their institutional interests?
2. **Competing interpretations:** What alternative readings of the operation are logically possible? What evidence would each interpretation require?
3. **Epistemic diagnosis:** Which interpretations does the visibility pipeline make easy to defend, and which does it render evidentially inaccessible? How does epistemic congruence vary across paradigms for this case?
4. **Paradigm consequences:** How did the case's passage through the pipeline affect broader theoretical debates? What became "known" as a result, and what remained invisible?

This structured approach enables systematic comparison across cases while remaining attentive to the historically specific pathways through which each operation became public knowledge.

4.5 Limitations

Three limitations warrant acknowledgment. First, the framework cannot determine whether dominant paradigms are substantively correct—only that their dominance cannot be straightforwardly attributed to empirical superiority. The visibility trap is an argument about evidential underdetermination, not a claim that institutional critique or competition paradigms are wrong.

Second, the analysis relies on the same public sources it critiques. We cannot access classified evidence that would allow direct assessment of coercion dynamics or operational intent. The contribution is therefore meta-theoretical: developing tools for reasoning about what we cannot know, rather than claiming privileged access to hidden truths.

Third, case selection is limited to operations attributed to state actors with substantial geopolitical significance. As noted in the scope conditions, the concealment mechanism plausibly extends to non-state and commercial actors—the financial sector being a clear example—but a developed account of the visibility trap in cybercrime, hacktivism, or operations by smaller states remains for future work.

5 Case 1: Stuxnet (2010)

The preceding sections developed the visibility trap framework in abstract terms. We now apply it to three canonical cases, tracing how each operation's passage through the pipeline shaped its paradigm-defining status. Stuxnet offers the clearest illustration of how high-congruence interpretations crowd out alternatives that would require inaccessible evidence.

5.1 Visibility story

In June 2010, sophisticated malware targeting Siemens industrial controllers was discovered, primarily in Iran. Analysis revealed Stuxnet was designed to sabotage centrifuges at the Natanz enrichment facility [7, 21, 47]. The public narrative emphasized unprecedented technical sophistication: four

zero-day exploits, stolen digital certificates, tailored sabotage logic. Attribution to a U.S.–Israeli operation emerged through leaks and investigative journalism [25].

5.2 Competing interpretations

Multiple hypotheses can explain Stuxnet, but they differ sharply in their evidentiary requirements and observability. The PETIO burden hypothesis—that the operation demonstrates the immense organizational investment required for high-impact cyber operations—requires evidence of sophisticated tooling and tailored targeting. This evidence is highly observable and abundantly available in vendor reports. Similarly, the trilemma/capability burn hypothesis—that Stuxnet illustrates how discovery degrades offensive capabilities—requires evidence of detection, patching, and subsequent operational degradation. Again, observability is high and the evidence is publicly documented.

The coercion hypotheses fare worse. If Stuxnet functioned as coercive signaling—a warning shot designed to compel Iranian restraint—we would need evidence of threats, demands, or controlled disclosure linking the operation to diplomatic objectives. Such evidence has low observability and is not publicly available. The strongest coercion claim—that Stuxnet achieved strategic delay by combining centrifuge degradation with diplomatic leverage—requires evidence connecting technical effects to bargaining outcomes. Observability here is very low, and no such evidence exists in the public record.

The asymmetry is stark: hypotheses aligned with institutional critique paradigms can marshal extensive public documentation, while coercion interpretations require evidence that the visibility pipeline cannot produce.

5.3 The coercion counterfactual

If Stuxnet were a calibrated coercive operation—designed to signal capability and resolve while compelling Iranian restraint—what would we expect to observe? Three categories of evidence would be necessary, and all three are structurally invisible to the pipeline. This parallels Trevorton’s insight about covert action: failures are visible, but successes—quiet compliance, deterred behavior—leave no trace [43].

First, **signaling architecture**: communications linking the operation to demands, backchannels confirming the coercer’s identity, timing coordinated with diplomatic pressure. If Washington signaled to Tehran through Swiss intermediaries that Stuxnet was a warning, we would never see it. Vendors can identify technical attribution indicators but cannot distinguish careless tradecraft from a deliberate calling card.

Second, **calibrated restraint**: evidence that attackers *could* have caused greater damage but chose not to. Stuxnet’s

narrow targeting is consistent with this interpretation—but restraint looks like incompetence or failure to external observers. An operation causing limited damage gets framed as “the attack underperformed,” not “the attacker intentionally pulled punches.”

Third, **target updating**: observable changes in Iranian behavior attributable to cyber pressure. If coercion works, the target complies quietly; success is a non-event. Iran has strong incentives to deny being coerced (signaling weakness invites further pressure). Any compliance may be delayed or indirect, making causal attribution ambiguous even to participants.

The asymmetry is structural. For PETIO/trilemma paradigms, success leaves traces (sophisticated tooling, eventual discovery, capability burn). For coercion, success is invisible: quiet compliance, no dramatic incident, no reason for the coercer to publicize. The visibility pipeline documents failure, not deterrence.

5.4 Epistemic consequence

Stuxnet became a paradigm anchor for Era 2 because its forensic story—sophistication, PETIO-level investment, capability burn—mapped onto institutional critique expectations. The visibility trap is that Stuxnet validated these claims *as an epistemic event*, because they predict traces vendors could document, not because coercion interpretations were falsified.

6 Case 2: NotPetya (2017)

6.1 Visibility story

On June 27, 2017, malware disguised as ransomware spread rapidly from Ukrainian networks to global systems. Technical analysis revealed NotPetya’s encryption was irreversible by design; its purpose was destruction, not extortion [20, 31]. Western governments attributed it to Russia, framing the global spillover—billions in damages to multinationals like Maersk and Merck—as collateral damage from an attack on Ukraine [44, 45].

6.2 The epistemic problem

The visibility pipeline made certain interpretations easy and others hard.

Easy (pipeline-congruent): PETIO burden—NotPetya drew on substantial investment, including prior tool reuse and supply-chain compromise. Trilemma dynamics—massive spillover is exactly what trilemma theory predicts when intensity is maximized.

Hard (pipeline-incongruent): Coercion interpretation—no visible threats, demands, or bargaining moves. Learning/testing interpretations—indicators do not surface publicly.

NotPetya is paradigm-shaping because the same pattern—severe spillover following high-intensity operation—is predicted both by a world where trilemma dynamics fundamentally drive cyber sabotage and by a world where only a subset of operations suffer such control failures. This is epistemic selection masquerading as empirical confirmation.

6.3 Alternative explanations and their traces

The trilemma interpretation—that Russia lost control of an operation intended to be more contained—is not the only reading consistent with public evidence.

Classified success despite visible failure. The spillover may have been tolerable collateral damage for an operation that achieved its actual objectives. If Russia's goal was degrading Ukrainian state capacity or signaling willingness to impose economic pain on sanctions-imposing countries, NotPetya succeeded. The “loss of control” framing assumes the attackers wanted containment; nothing in the public record establishes this.

Punitive attack designed to appear uncontrolled. Deniability is enhanced when an operation looks like a mistake rather than deliberate punishment. The same forensic pattern—rapid propagation, indiscriminate targeting—is produced by genuine control failure and by deliberate design mimicking control failure. From public evidence, these are indistinguishable. This is the logic of strategic denial and deception: a deliberately engineered appearance of accident is a recognized instrument of statecraft, and the technical record cannot by itself separate authentic loss of control from its skilled simulation [13].

Idiosyncratic failure, not systemic dynamic. NotPetya may reflect organizational pathologies specific to GRU Unit 74455, not an iron law of cyber operations. Distinguishing idiosyncratic from systemic failure would require a denominator: how many high-intensity operations *did not* produce catastrophic spillover? That denominator is unknowable from public sources.

6.4 Epistemic consequence

NotPetya reinforced Era 2 by providing an apparent smoking gun for trilemma claims. But the same evidence supports multiple interpretations, and the trilemma's dominance reflects its congruence with visible spillover, not demonstrated superiority over alternatives.

7 Case 3: SolarWinds (2020)

7.1 Visibility story

In December 2020, FireEye revealed it had been breached via compromised SolarWinds Orion software [8]. Investigation uncovered supply-chain compromise, selective second-stage payloads, exceptional operational security, and months-long dwell time. U.S. agencies attributed the campaign to Russian intelligence, describing it as espionage [42].

7.2 The epistemic problem

Competition paradigm interpretations were highly congruent: extraordinary stealth, long dwell time, and campaign framing fit theories emphasizing intelligence collection and positional advantage.

Yet the same pattern is consistent with other logics: the attackers may have established access for future sabotage or coercive leverage; mixed motives could have pursued intelligence while creating disruption options. Stealthy, persistent access is a hallmark of espionage but also a prerequisite for coercive or destructive operations. Public evidence cannot discern whether SolarWinds was “only” espionage.

7.3 Political economy of interpretation

The sophisticated espionage interpretation served multiple actors. For FireEye, highlighting sophistication demonstrated detection capabilities. For the U.S. government, framing it as intelligence-gathering made it a manageable problem. For victims, this narrative emphasized an unusually capable adversary over routine security failure. For academia, it fit competition paradigm theories.

Alternative interpretations—that the operation failed its objectives or prepared disruption that never materialized—lacked institutional champions and would require classified evidence.

7.4 Epistemic consequence

SolarWinds elevated competition paradigms because institutional actors had incentives to interpret an ambiguous operation as sophisticated espionage. This elevation is itself part of the visibility trap: ambiguous cases congruent with a paradigm's observable signatures are claimed as validation.

8 Negative cases: WannaCry and Sony

The preceding cases illustrate the visibility trap in action, operations that became paradigm-defining because their observable features aligned with high-congruence frame-

works. But if visibility alone determined paradigm influence, any well-documented incident should shape strategic theory. The following cases test this implication by examining incidents with high visibility that nonetheless failed to achieve canonical status.

The framework's explanatory power is clarified by examining cases that *did not* become paradigm-defining despite substantial visibility (Table 2).

8.1 WannaCry (2017): visibility without paradigm shift

In May 2017, WannaCry ransomware spread globally, affecting over 200,000 systems across 150 countries. Attribution to North Korea emerged through government statements [45]. WannaCry exhibited massive spillover—features that should have reinforced institutional critique paradigms. Yet it did not achieve paradigm-shaping status.

Several factors explain this. **Attribution to an outlier state:** North Korea is viewed as reckless and motivated by financial desperation rather than strategic calculation, making WannaCry “exceptional” rather than theoretically generative. **Mixed criminal/state framing:** The ransomware mechanism complicated strategic interpretation. **Accidental termination:** The kill-switch activation muddled causal inference.

WannaCry demonstrates that visibility is necessary but not sufficient for paradigm influence. The *attribution frame*—how the responsible actor is categorized—mediates whether visible evidence gets incorporated into mainstream strategic theory.

8.2 Sony pictures (2014): coercion made visible

In November 2014, attackers breached Sony Pictures, destroying data, leaking information, and issuing threats linked to the film *The Interview*. The operation included explicit demands, public threats, and partial compliance. The U.S. government attributed it to North Korea.

Sony exhibited precisely the features coercion paradigms require: explicit demands, clear attribution, observable target response. Yet it did not rescue the coercion paradigm. **Non-state target:** Sony was a corporation, not a state; coercing a private company did not fit the paradigm's core domain. **Attribution to North Korea:** DPRK's outlier status reduced generalizability. **Ambiguous success:** Sony initially withdrew the film, then reversed course.

Sony reveals that epistemic congruence is domain-specific. Paradigms define not only what evidence counts but what *cases* count. The visibility trap operates through case-selection criteria that precede evidence evaluation. Sony also illustrates the deterrence–compellence asymmetry developed above, and in a way the framework predicts rather

than merely accommodates. The reason Sony displayed “precisely the features coercion paradigms require” is that it was an act of *compellence*: it carried an explicit demand (suppress the film), a visible deadline, and an observable act of compliance [38]. Compellence is, by Schelling's logic, the observable half of coercion. The operations that would vindicate the coercion paradigm in the register of Stuxnet or NotPetya, by contrast, would be *deterrent*—warnings and quiet inducements whose success is a non-event. That the single coercion case to achieve real visibility was the compellent one is exactly what the visibility trap implies: the paradigm is not absent from the record because it fails, but because its characteristic successes take the deterrent, footprint-free form, while its rare visible instances are dismissed on other grounds (here, a corporate target and an outlier state). Sony is thus not an awkward exception to the framework but a demonstration of its central mechanism.

9 Synthesis: the visibility trap in action

The case analyses reveal a consistent pattern: operations become paradigm-defining not primarily because they settle theoretical debates but because their passage through the visibility pipeline makes certain interpretations documentable while rendering others inaccessible. This section consolidates these findings into a causal mechanism, derives testable hypotheses, and addresses alternative explanations.

9.1 The causal mechanism: from congruence to dominance

The visibility trap operates through a four-stage causal sequence:

Stage 1: Pipeline selection. The visibility pipeline admits a biased sample of operations, those that failed visibly, targeted monitored networks, or served political attribution purposes. This sample systematically over-represents operations whose observable features align with high-congruence paradigms.

Stage 2: Institutional amplification. Evidence producers have incentives that reinforce congruence bias. Vendors emphasize sophistication and persistence (validating their services). States attribute when politically useful. Victims emphasize capable adversaries over security failures. Academia favors cases fitting recognized debates. Crucially, no institutional actor benefits from generating evidence that would undermine dominant paradigms.

Stage 3: Terminological capture. Vendors shape the conceptual infrastructure through which operations are understood. Terms like “Advanced Persistent Threat,” “supply-chain compromise,” and “dwell time” are vendor-created

Table 2 Why some visible cases became paradigm-defining

Case	Visibility	Paradigm impact	Mediating factors
Stuxnet	High	Era 2 anchor	State target; “normal” actors; strategic stakes
NotPetya	Very High	Era 2 reinforced	State-linked targets; Russia as strategic actor
SolarWinds	High	Era 3 elevated	Government victims; great-power rival
WannaCry	Very High	Limited	DPRK as outlier; criminal/state ambiguity
Sony	High	Minimal	Corporate target; DPRK as outlier

categories that embed competition-paradigm assumptions [3]. When vendors name and track “APT29” or “Fancy Bear,” they construct entities whose persistence becomes a fact of the strategic landscape. Certain paradigms have ready-made vocabularies; others must work against the grain of established terminology.

Stage 4: Canonization and path dependence. Cases passing through the pipeline with aligned interpretations become canonical examples. Once canonized, paradigms shape what questions get asked and what evidence gets sought. Low-congruence paradigms struggle to generate confirming cases not only because their evidence is harder to observe but because researchers stop looking.

9.2 Testable hypotheses

A theory-building contribution is only as valuable as the empirical work it enables. The visibility trap framework generates several observable implications that distinguish it from alternative explanations and provide concrete targets for subsequent research. If epistemic congruence shapes paradigm dominance as theorized, the following patterns should be detectable in citation data, doctrinal texts, attribution practices, and vendor reporting:

H1 (Citation patterns): Post-Stuxnet, publications emphasizing PETIO/trilemma concepts should show increased citation rates relative to coercion-focused work.

H2 (Doctrine analysis): U.S. cyber strategy documents should show increasing competition-paradigm language post-2016, with institutional critique language prominent post-2010.

H3 (Attribution patterns): Incidents receiving prompt public attribution should disproportionately be those congruent with prevailing paradigms.

H4 (Paradigm sponsorship): Vendors should disproportionately frame incidents in ways emphasizing sophistication and persistence; government attributions should frame incidents in ways supporting current strategic priorities.

9.3 Addressing alternative explanations

Great-power rivalry intensification. If rivalry alone drove paradigm adoption, coercion paradigms—equally consistent with competition narratives—should have gained traction. They did not. The differential adoption of competition over coercion frameworks suggests visibility/congruence is doing independent work.

Genuine disconfirmation. The framework’s claim is that we cannot tell from public evidence whether coercion fails or merely fails to be documented. Genuine disconfirmation and observational absence look identical from outside.

Capability evolution. Capability changes are real, but should affect all paradigms proportionally. The differential success of high-congruence paradigms requires an additional explanation. Epistemic congruence provides it.

10 Implications and conclusion

10.1 The stakes

If our read of the threat landscape really is tilted toward whatever the pipeline makes visible, the costs are not just academic. Defensive money and attention may pour into resilience against the spectacular, headline-grabbing failure while the quieter, more controlled operation goes unbudged. Doctrine may keep overlooking coercive options simply because they sit below the visibility threshold and so never enter the evidence base. And any strategic assessment that works by aggregating visible incidents will, almost by design, misdescribe what adversaries are actually doing—because the visible incidents are not a fair sample.

10.2 Prescriptions

For scholars, the practical upshot is to stop treating vendor reports and public attributions as raw fact and start treating them as artifacts—shaped by commercial and political interest—and to chase the sources that partly escape the pipeline: leaks, interviews, declassified files. A good theory of cyber strategy has to double as a theory of its own evidence.

For practitioners, the caution is against drawing strong conclusions from the public cases alone, and toward asking a harder question: is our posture tuned to the threats we can see, or to the full range of what an adversary can actually do?

For doctrine, it is simply to hold the possibility that the frameworks we have inherited describe what is visible better than they describe what matters.

10.3 Conclusion

This paper has set out a way of thinking about how the visibility trap shapes what we take ourselves to know about cyber strategy. The pipeline turns clandestine operations into public evidence through a series of stages that quietly favour some paradigms over others, and epistemic congruence is what predicts which frameworks end up looking confirmed and which end up running doctrine.

None of this shows the dominant paradigms are wrong. Institutional critique and competition may well be capturing something real. The point is narrower and, we think, harder to shake: we cannot easily tell whether they dominate because they are right or because they are congruent, since the public record would look much the same either way. That is also where this account departs from existing work. Saying cyber coercion is hard is not new; the claim here is that we are not in a position to know how hard it is, because the evidence that would tell us is precisely the evidence the pipeline does not produce—which means the familiar verdict against coercion may be measuring our instruments rather than the world.

Put in the language of agnotology, the pipeline manufactures structured ignorance—blind spots held in place by the very machinery that generates what we do know. Put in the language of intelligence studies, it carries into the cyber domain a lesson covert-action scholars learned long ago: we study a biased sample, and the way cases become visible may be tangled up with the very things we are trying to explain. What we are left with is a canon that may say more about the political economy of cyber forensics than about what drives states in cyberspace—and noticing that is the first step toward a more modest, and more useful, picture of cyber strategy.

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